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An ultra-lightweight design for imperceptible plastic electronics

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Electronic devices have advanced from their heavy, bulky origins to become smart, mobile appliances. Nevertheless, they remain rigid, which precludes their intimate integration into everyday life. Flexible, textile and stretchable electronics are emerging research areas and may yield mainstream technologies[1–3]. Rollable and unbreakable backplanes with amorphous silicon field-effect transistors on steel substrates only 3 μm thick have been demonstrated[4]. On polymer substrates, bending radii of 0.1 mm have been achieved in flexible electronic devices[5–7]. Concurrently, the need for compliant electronics that can not only be flexed but also conform to three-dimensional shapes has emerged[3]. Approaches include the transfer of ultrathin polyimide layers encapsulating silicon CMOS circuits onto prestretched elastomers[8], the use of conductive elastomers integrated with organic field-effect transistors (OFETs) on polyimide islands[9], and fabrication of OFETs and gold interconnects on elastic substrates[10] to realize pressure, temperature and optical sensors[11–14]. Here we present a platform that makes electronics both virtually unbreakable[4] and imperceptible. Fabricated directly on ultrathin (1 μm) polymer foils, our electronic circuits are light (3 g m^{-2}) and ultraflexible and conform to their ambient, dynamic environment. Organic transistors with an ultra-dense oxide gate dielectric a few nanometres thick formed at room temperature enable sophisticated large-area electronic foils with unprecedented mechanical and environmental stability: they withstand repeated bending to radii of 5 μm and less, can be crumpled like paper, accommodate stretching up to 230% on prestrained elastomers, and can be operated at high temperatures and in aqueous environments. Because manufacturing costs of organic electronics are potentially low, imperceptible electronic foils may be as common in the future as plastic wrap is today. Applications include matrix-addressed tactile sensor foils for health care and monitoring, thin-film heaters, temperature and infrared sensors, displays[15], and organic solar cells[16].

An example of an ultrathin active-matrix array with resistive tactile sensors is shown in Fig. 1a. The electronic circuit foil weighs so little that it floats through air much like a feather (Fig. 1b and Supplementary Movie 1). At only 3 g m^{-2}, our sensor foil is 60-fold lighter than conventional polyimide substrates (125 μm thick, 180 g m^{-2}), 27-fold lighter than office paper (130 μm thick, 80 g m^{-2}) and still eightfold lighter than a steel foil 3 μm thick (24.2 g m^{-2}). The unprecedented mechanical properties of the foil are illustrated in Fig. 1c, in which it survives being crumpled like a piece of paper. These imperceptible ultrathin foils form an intimate contact with arbitrary shapes (including human skin) without imposing constraints either mechanically or by weight. In the ultrathin array in Fig. 1, a polyethylene naphthalate (PEN) foil 1 μm thick was used as substrate; the circuits were formed with a hybrid anodic aluminium oxide[17,18] and a phosphonic acid self-assembled monolayer[19] (SAM) gate dielectric, dinaphtho[2,3-b:2′,3′-f]thieno[3,2-b]thiophene (DNTT) as air-stable organic semiconductor[20], and a parylene interlayer separating the active-matrix backplane from the sensing layer. The fabrication of the device consists exclusively of process steps commonly used in the semiconductor industry, such as vacuum evaporation (metals and organic semiconductor) and chemical vapour deposition (parylene). Potentiostatic anodization was used to form ultra-dense and durable high-*k* dielectric layers a few nanometres thick for the organic transistors[17]. The self-healing nature of the anodic oxide enables us to fabricate complex devices as thin as 100 nm on industrial-scale plastic substrates with considerable (88 nm root mean square) surface roughness, while maintaining high performance (see Methods and Supplementary Information for details). Typically, all 12 × 12 sensor cells on an active-matrix array are functional.

Figure 2 shows potential applications of our technology as an infrared bolometer sensor (Fig. 2a, b) and in heat management, where high electrical power is effectively dissipated to a heat sink (Fig. 2c–j). For these applications, a thin gold metal film 100 nm thick was evaporated onto the 1-μm PEN foil. The transient temperature signals in Fig. 2b show temperature equilibration within 9 μs after excitation with a 10-ns laser pulse from an Nd:YAG laser at 1,064 nm wavelength. The bolometer has a small thermal mass of $C = 8\ \mu J\ K^{-1}$, comparable to micromachined devices[21,22]. Fast temperature equilibration allows efficient power dissipation from the device to an underlying substrate, thus enabling efficient heat management strategies in flexible and stretchable electronics, an emerging topic[23,24]. As depicted in the setup photograph and in the series of infrared thermal images in Fig. 2c–j, the electronic foil dissipates an electrical power of 0.35 W to a Teflon substrate and a 20-fold higher power to a heat sink substrate with current densities comparable to those found in integrated circuits[25]. An example of its use as a clinging temperature sensor, wrapped around a capillary of a high-performance liquid chromatography pipe, is shown in Supplementary Fig. 1. Localized heating, as in the example of a thin metal-foil sensor for temperature sensing and heating on a human finger, is demonstrated in Supplementary Fig. 2a, b.

We added complexity to the devices by fabricating thin-film transistors for actively controlled sensors and actuator networks. Aluminium serves as the gate electrode, potentiostatically anodized aluminium oxide is the gate dielectric, an n-tetradecyl phosphonic acid SAM provides the interface to the air-stable high-mobility p-type organic semiconductor DNTT. Gold source and drain contacts complete the device (Fig. 3a). An optical micrograph of a single transistor on the 1.2-μm PEN substrate (channel length 40 μm, width 500 μm) is shown in Fig. 3b. A detailed analysis of the surface morphology by atomic force microscopy is provided in Supplementary Fig. 3. For mechanical testing devices were encapsulated with a parylene layer 0.8–1 μm thick that places the thin-film transistor near the neutral plane position[5,6,8] (see Methods). A cross-sectional scanning electron microscopy (SEM) image of an encapsulated device with a total thickness of 2 μm is shown in Fig. 3c. The layer structure of the 100-nm OFET is revealed in the scanning transmission electron microscope (STEM) cross-sectional image (Fig. 3d). The amorphous nature of the aluminium oxide layer

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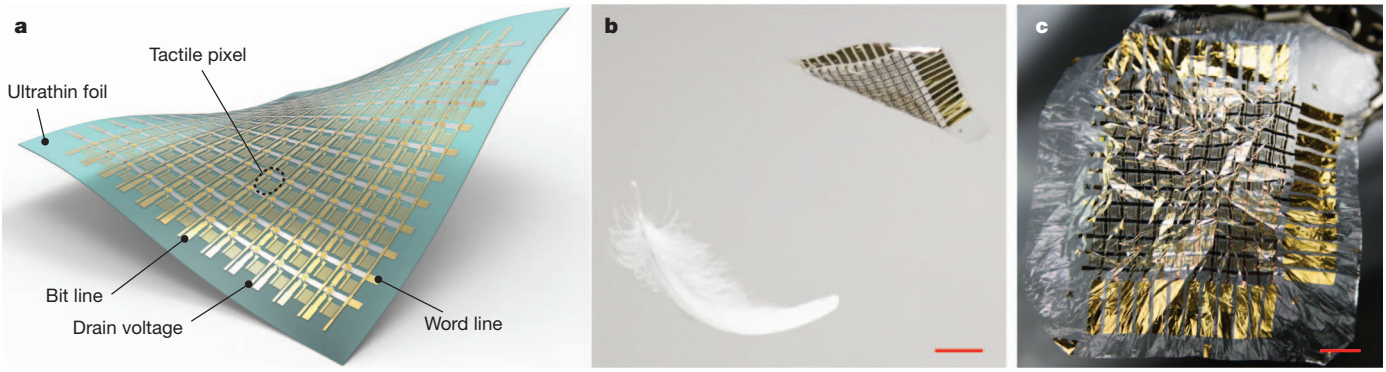

Figure 1 | Imperceptible electronic foil. **a**, Illustration of a thin large-area active-matrix sensor with 12 × 12 tactile pixels. **b**, Ultrathin plastic electronic foils are extremely lightweight (3 g m^{-2}); they float to the ground more slowly than a feather and are therefore virtually unbreakable. Scale bar, 2 cm. **c**, At only 2 μm thickness, our devices are ultraflexible and can be crumpled like a sheet of paper. Scale bar, 1 cm.

is confirmed by the absence of crystalline regions in the bright-field STEM image in Supplementary Fig. 4a. The elemental composition of the individual layers was analysed by means of an energy-dispersive X-ray system (Supplementary Fig. 4b, c), which confirmed the formation of a SAM layer. We then compared our anodic aluminium oxide hybrid dielectric with conventional aluminium oxide prepared by treatment with oxygen plasma (Supplementary Fig. 5). In short, potentiostatically anodized alumina shows higher mechanical durability than plasma-formed alumina, and withstands large bending strains at relevant operating voltages. This, together with the ultrathin architecture, allows our transistors to survive crumpling like a sheet of paper without performance degradation (Supplementary Fig. 6). Transfer and output characteristics of an OFET with a saturation mobility of 3 cm^2 V^{-1} s^{-1} are plotted in Fig. 3e, f; Supplementary Fig. 7 characterizes devices with 9 nm gate oxide. The devices have a performance equal to that of reference devices fabricated on rigid, smooth Si/SiO_2 substrates (Supplementary Fig. 8), and are on a par with the best-performing OFETs reported so far[26].

To demonstrate the potential of our technology, we fabricated a 12 × 12 active-matrix array with integrated resistive tactile sensors. The diagram in Fig. 4 shows the design of a tactile pixel. The control transistors were fabricated on the 1.2-μm substrate with a high aspect ratio (channel width-to-length ratio 12,000 μm/20 μm), which makes the active-matrix backplane suitable for use in high-power applications, for instance as OLED pixel drivers. After the formation of an insulating interlayer of 800-nm parylene, interdigitated gold electrodes were deposited as resistive touch sensors. The electrical connection between the touch sensors and the switching matrix backplane is provided by gold evaporated through high-resolution vias drilled with a green laser marker system. The rough surface of the parylene interlayer drastically improves the adhesion of the gold sensor layer, providing a robust and wear-resistant tactile sensing pixel (Supplementary

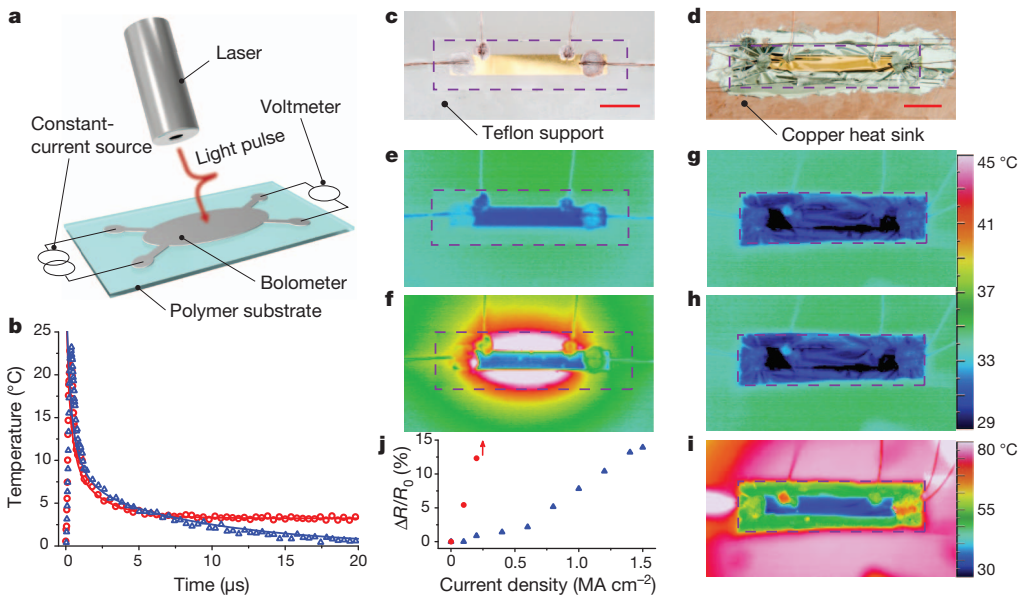

Figure 2 | Thin-film infrared sensor and heat management. **a**, Diagram of a thin-film infrared bolometer sensor and experimental setup to measure the transient temperature signal on excitation by a 1,064-nm Nd:YAG laser pulse. **b**, Temperature change after thermal pulse excitation of a free-standing bolometer (red circles) and a bolometer coupled to a heat sink (blue triangles). Heat is transferred to the heat sink very rapidly (on a timescale of only $4\tau = 9$ μs). Solid lines are fits according to equations (1) and (2) in Supplementary Information. **c–j**, Heat management in thin-film electronics. Metallic conductors (100-nm Au) on a 1.2-μm PEN foil are placed on a Teflon support with low thermal conductivity (**c**) and on a copper block serving as heat sink (**d**). Scale bars, 2 mm. The excellent heat transport properties of such a thin-foil substrate allow these devices to handle the current densities used in integrated circuits and to dissipate a significant amount of power when thermally connected to a heat sink. Thermography images visualize dissipated heat on Teflon at zero current (**e**) and 0.2 A (**f**) and the copper support at zero current (**g**), 0.2 A (**h**) and 1.5 A (**i**). **j**, Resistance change $\Delta R/R_0$ of a conductor on Teflon support (red circles) and in thermal contact with a copper heat sink (blue triangles). While Teflon supported conductors fail at a current density of 0.2 MA cm^{-2}, a current density tenfold higher is sustained when the device is placed on a heat sink.

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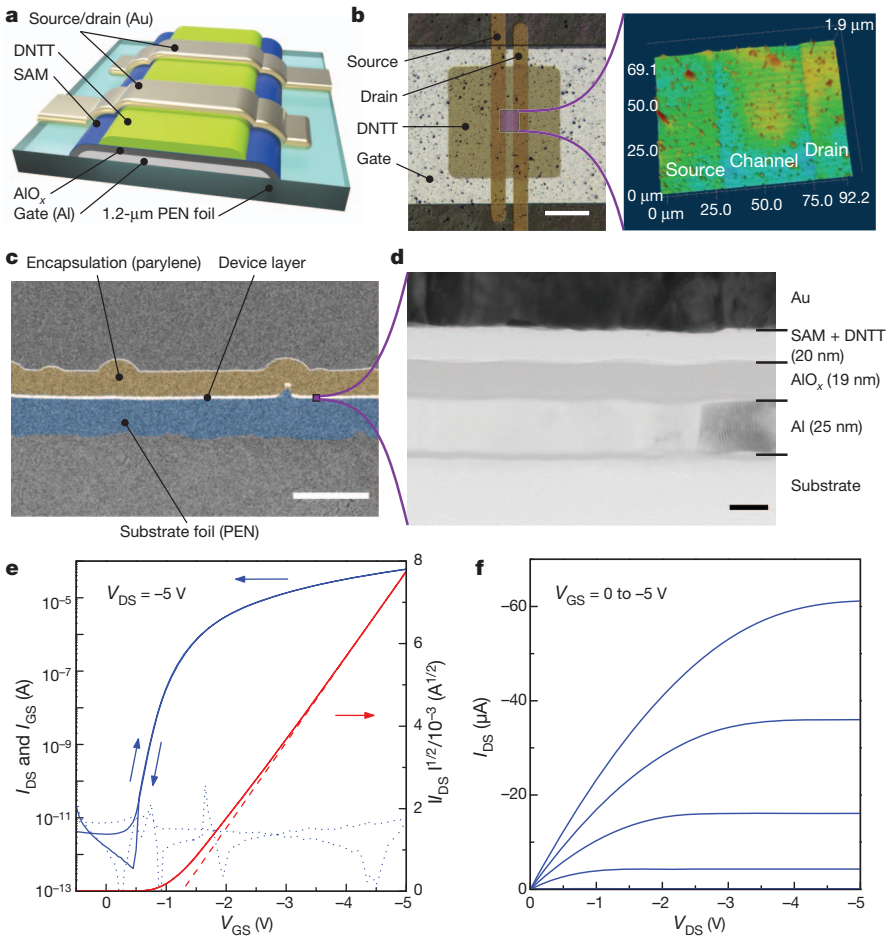

Figure 3 | Thin-film transistors on 1.2-μm PEN foils. **a**, Cross-sectional diagram of an ultrathin OFET. Aluminium serves as gate contact and a mechanically durable and dense hybrid dielectric is formed from potentiostatically anodized alumina and a phosphonic acid SAM. The organic semiconductor DNTT ensures air-stable operation, and 50 nm thick Au serves as the source and drain. The complete device including the PEN substrate is 1.3 μm thick. **b**, Optical micrograph of a single transistor on the 1.2-μm PEN substrate with a channel length of 40 μm and width of 500 μm. Scale bar, 200 μm. The magnified three-dimensional view of the channel region highlights the topology of the transistor on the unconventionally rough substrate. **c**, Cross-sectional SEM image of the ultrathin transistor encapsulated with 800-nm parylene diX-SR to protect against mechanical damage and place the transistor near the neutral strain position. The total thickness of this device is about 2 μm. Scale bar, 2 μm, **d**, High-resolution cross-sectional TEM image showing the layer structure of the 100-nm OFET. The amorphous aluminium oxide layer 19 nm thick ensures low leakage currents even on rough substrates and is mechanically reliable, while maintaining low operating voltages. An n-tetradecyl phosphonic acid SAM about 2 nm thick forms the interface to a DNTT layer roughly 18 nm thick. Scale bar, 20 nm. **e**, Transfer characteristics of an ultrathin, flexible OFET. Plotted are drain-source current (I_{DS}) and gate-source current (I_{GS}) as a function of gate-source voltage (V_{GS}) at a drain-source voltage (V_{DS}) of −5 V. The saturation mobility is 3 $cm^2 V^{-1} s^{-1}$, the on/off ratio exceeds seven orders of magnitude and the gate leakage current (I_{GS}) is less than 10 pA. **f**, Corresponding output characteristics. Plotted are I_{DS} as a function of V_{DS} for a gate-source voltage from 0 V to −5 V.

Fig. 9). Figure 4b shows a top-view photograph of the completed 8 cm × 8 cm ultrathin tactile sensor array; a magnified view of four pixels is shown in the inset. A detailed view of a single tactile pixel is shown in Fig. 4c, together with a single backplane transistor without the sensor layer and a magnified view of the transistor's active region. A circuit diagram of the resistive sensor pixel illustrates the operating principle of the sensor cell. Figure 4d plots the transfer characteristic of a transistor in the active-matrix configuration (without the sensor layer). Our transistors maintain saturation mobilities of up to 1.6 $cm^2 V^{-1} s^{-1}$ when operated at only 3 V source drain and gate voltage and an on/off-current ratio of eight orders of magnitude. The ultra-dense hybrid dielectric layer ensures low gate leakage currents of less than 100 pA even for transistors with a high channel width-to-length ratio. Using a high width-to-length ratio enables output currents of more than 0.5 mA at 3 V, which is sufficient for OLED display drivers and power amplifiers. The average mobility across all 144 transistors on the tactile sensor foil is 0.88 ± 0.13 $cm^2 V^{-1} s^{-1}$, as the mobility distribution plotted in Fig. 4e shows. A critical design parameter for active-matrix-based sensor arrays is the on/off ratio of the control transistors, which must be large enough to allow unambiguous distinction between on and off pixels. The on/off ratio for all of the 144 pixels of our ultrathin tactile sensor exceeds 10^7 (on average $(2.57 \pm 0.7) \times 10^7$; see distribution in Supplementary Fig. 10f). Threshold voltage, sub-threshold swing and on-current and off-current distributions are shown in Supplementary Fig. 10. When connected, the drain current of a tactile sensing pixel depends on the surface conductivity of the sensed object. Figure 4f shows the transfer curve for a single pixel when the interdigitated resistive sensor is in contact with various objects. Highly conducting surfaces are readily detected, and even non-conducting objects can be perceived when an anisotropically conducting pressure-sensitive rubber is laminated to the tactile sensing array (a soft touch with 1 kPa pressure is readily registered). Such tactile sensing foils can be used as human–machine interfaces, because the resistivity of human skin is well suited to activating a pixel with a gentle touch. More details and high-resolution images of the individual fabrication steps are given in Supplementary Fig. 11. To demonstrate the functionality as a tactile sensor, we recorded the position of a metallic ring when placed on the sensing sheet. Figure 4g shows a top-view photograph of the ring and the corresponding signal map. The circular pattern is read out with high precision, and a difference of more than six orders of magnitude in the on-current between contacted and non-contacted pixels is observed. Such ultrathin large-area sensing sheets readily conform to three-dimensional surfaces, allowing unprecedented applications: displays, sensors, thermal imaging in sports and recreation, medical treatment and prosthetics; electronic skin; robotics; and consumer electronics. The example of a tactile sensor conforming to a model of the upper human jaw (Fig. 4h) illustrates potential applications in health care and monitoring, in which imperceptible sensing and actuating systems are required to minimize patient discomfort. In biomedical applications, including *in vivo* sensing of biosignals, ultrathin electronic foils must operate at elevated temperatures and in aqueous environments. At only 2 μm thickness (including parylene encapsulation), our transistors can withstand thermal annealing at up to 170 °C (Supplementary Fig. 12) and operate continuously for more than two weeks fully immersed in saline solution without significant degradation in performance (Supplementary Fig. 13).

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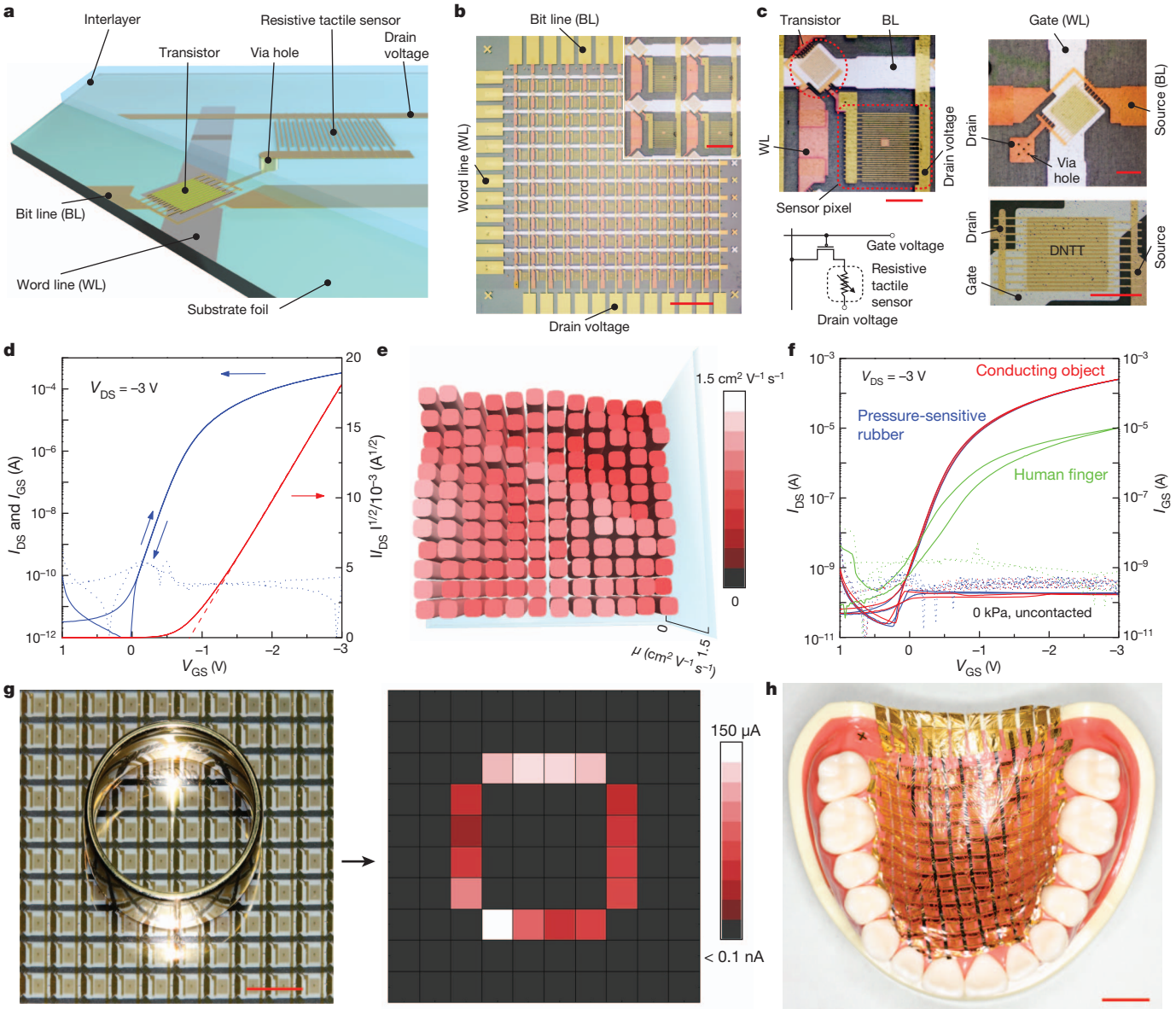

Figure 4 | Active-matrix tactile sensing foil. **a**, Design of a tactile pixel comprising a switching transistor and a resistive touch sensor. The control transistor is fabricated directly on the 1.2-μm substrate. An interlayer of 800-nm parylene insulates the switching matrix backplane from the sensing layer. The electrical connection to the transistors is provided through high-resolution vias. Interdigitated gold electrodes with a line pitch of 30 μm serve as touch sensors. **b**, Top-view photograph of the completed 8 cm × 8 cm ultrathin tactile sensor array with 12 × 12 pixels. Scale bar, 1 cm. A magnified view of four pixels is shown in the inset. (Scale bar: 2 mm) **c**, Top: detailed view of an individual tactile pixel together with a single backplane transistor (left) and without the sensor layer (right). Scale bars, 1 mm. Bottom: circuit diagram of the resistive sensor pixel (left) and magnified view of the transistor's active region (right). Scale bar, 500 μm. **d**, Transfer characteristic of a transistor in the matrix configuration. The saturation mobility is 1.6 $cm^2 V^{-1} s^{-1}$ at 3 V source-drain and gate voltage, on/off-current ratio is 10^8. The ultra-dense dielectric layer ensures low gate leakage currents (less than 100 pA), enabling output currents of more than 0.5 mA at 3 V. **e**, Typical mobility distribution of all 144 transistors on a completed tactile sensor. The average mobility is 0.88 ± 0.13 $cm^2 V^{-1} s^{-1}$. **f**, Transfer curve for a single pixel when the interdigitated resistive sensor is touched by various objects. **g**, Top-view photograph (left) and corresponding drain current of a metallic ring placed on the sensing sheet (right). A difference of more than six orders of magnitude is observed in drain current between contacted and non-contacted pixels. Scale bar, 1 cm. **h**, Tactile sensor sheet tightly conforming to a model of the human upper jaw. Scale bar, 1 cm.

To demonstrate the stretch compatibility of our lightweight electronic foils, we laminated devices to a pre-stretched elastomeric film. When the pre-stretched elastomer is allowed to relax, wrinkles form in the active transistor layer, which enables stretching of the device in the direction of the initial deformation of the elastomer[3,27–30]. The electronic circuits were fabricated close to the neutral plane of the foil stack, which permits bending to a radius of as little as 5 μm with marginal change in electrical performance. Figure 5a illustrates how stretchability of ultrathin electronics is achieved: after fabrication of the devices on the carrier substrate, the electronic film is peeled off the support stack and laminated to a pre-stretched elastomer (3M VHB) by simple picking and placing. When the prestrain in the elastomer is relaxed, adhesion between the elastomer and the thin electronic foil causes the strain in the elastomer to transfer to the thin foil, forming a network of out-of-plane wrinkles in the device that accommodates subsequent tensile strains. Figure 5b shows a picture sequence of this formation of folds when the elastomer is relaxed. The wrinkle topology was characterized by both three-dimensional optical microscopy and SEM (Fig. 5c). To determine the minimum bending radii of the multiple folds that form in the active area of the transistors, we analysed cross-sectional SEM images of a transistor undergoing 233% tensile strain. Figure 5d reveals multiple folds with bending radii of less than

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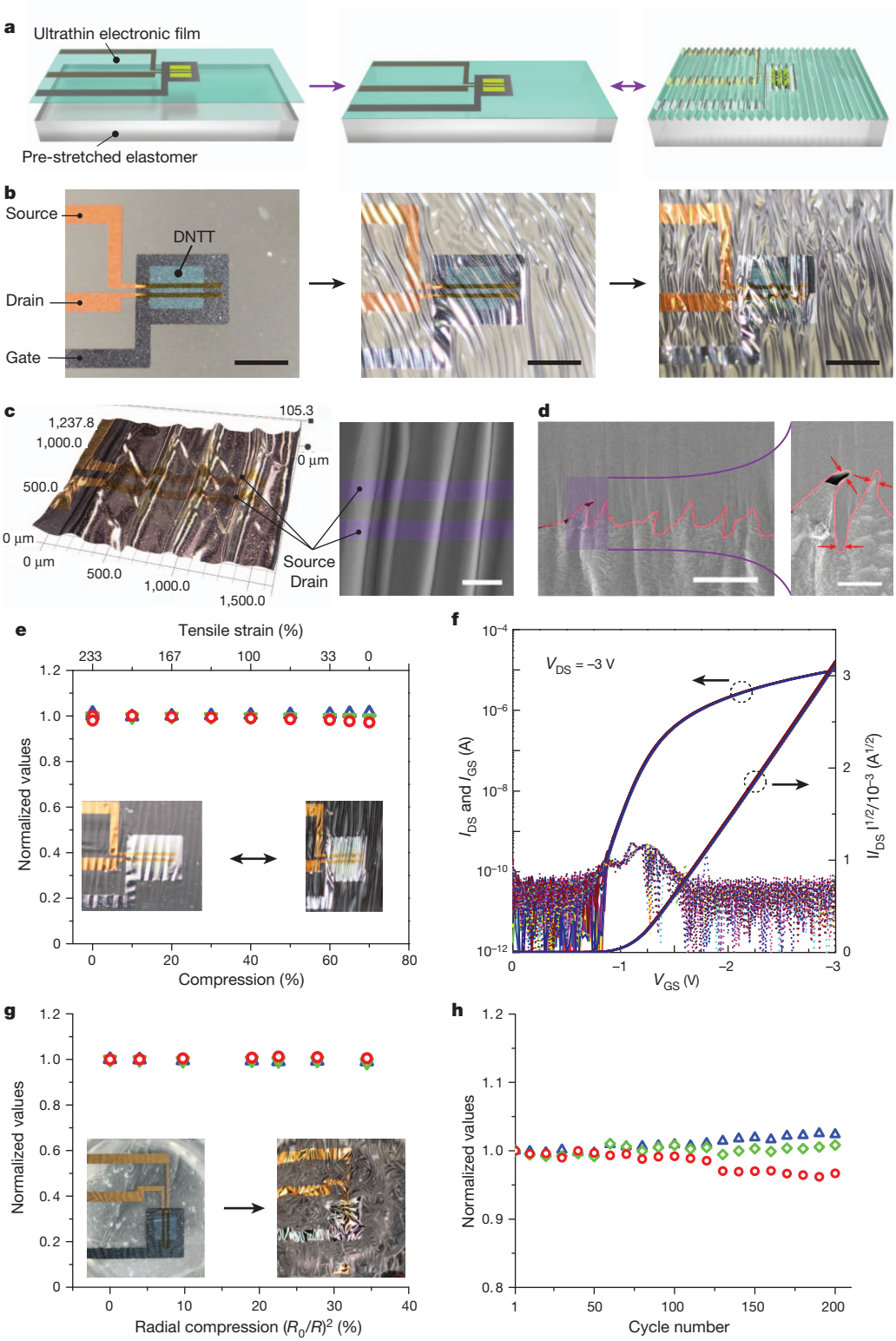

Figure 5 | Stretch-compatible ultraflexible transistors. **a**, Illustration of stretchable ultrathin electronics. After fabrication, devices are peeled off the support stack and laminated to a pre-stretched elastomer (3M VHB) by simple picking and placing. When the prestrain in the elastomer is relaxed, the ultrathin electronic foil forms a network of out-of-plane wrinkles that accommodate subsequent tensile strains. **b**, Picture sequence of the formation of folds in an ultraflexible transistor when the elastomer is relaxed. Scale bars, 1 mm. **c**, Wrinkle topology characterized by a three-dimensional optical microscope (left) and by SEM (right). The out-of-plane folds are about 100 μm high. Scale bar, 200 μm. **d**, Cross-sectional SEM image of a transistor undergoing 70% compressive strain. The magnified view reveals multiple folds with bending radii of less than 5 μm within the active device area. Scale bars, 100 μm (left); 20 μm (right). **e**, On-current (red circles), mobility (green diamonds) and threshold voltage (blue triangles) as functions of compressive and tensile strains. No irreversible degradation was observed during the experiment. Inset: a transistor in relaxed (left) and compressed (right) states. **f**, Transfer curves corresponding to **e**, demonstrating the mechanical stability of the hybrid gate dielectric: no increase in gate leakage current was observed. **g**, Biaxially stretchable transistor. When laminated to an equi-biaxially prestretched elastomer, these transistors withstand a 35% decrease in area before failure. The inset shows the transistor in relaxed (left) and compressed (right) states. **h**, Mechanical durability during repeated compression and re-stretching to 100% tensile strain. Plot of on-current (red circles), mobility (green diamonds) and threshold voltage (blue triangles) against stretch cycles. Even after 200 full cycles, less than 4% change in device parameters was observed.

5 μm within the active device area. Despite these extreme bending conditions, the transistors not only remain fully functional, but, as a result of the neutral-plane fabrication, there is virtually no performance degradation even at 233% tensile strain. Figure 5e shows on-current, mobility and threshold voltage as functions of compressive and tensile strains. The maximum compression is determined by the pre-strain of the elastomer and the packing density of the out-of-plane folds. No irreversible degradation was observed during the experiment. The corresponding transfer curves (Fig. 5f) demonstrate the superior mechanical stability of our hybrid gate dielectric: no increase in gate leakage current was observed. In a biaxially pre-stretched elastomer, a random network of wrinkles forms on relaxation. These devices remain fully functional up to a 35% decrease in area (Fig. 5g); beyond this value, the bending strain on the active device becomes too great and dielectric breakdown occurs. Because no measures were taken to control the formation of wrinkles, we expect selective bonding of the electronic foil to the pre-strained elastomer to further improve performance in this area. Finally, we explored the mechanical durability of uniaxially

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stretchable devices by repeated compression and re-stretching to 100% tensile strain. An image sequence taken during a stretching cycle is shown in Supplementary Fig. 14 and Supplementary Movie 2. Even after 200 full cycles the change in device parameters amounts to less than 4% (Fig. 5h), which—estimated conservatively—suggests a t_{80} lifetime of more than 1,000 full cycles (t_{80} is the number of cycles until device parameters degrade to 80% of their initial values). More information on the stretching test setup and additional experimental results is given in Supplementary Figs 15–17.

The technology described here is based on common organic electronic manufacturing methods, and enables the fabrication of imperceptible electronic systems containing integrated circuits, sensors and power supplies. At only 2 μm total thickness, these electronic devices are extremely lightweight and minimize electronic waste; like plastic wrap, they are easily applied to curvilinear and dynamic surfaces and can therefore be used in consumer and mobile electronic appliances, architectural design, robotics, emergency response, sports, healthcare and biomedical systems.

METHODS SUMMARY

PET sheets (thickness 125 μm) covered with a thin polydimethylsiloxane layer served as supporting substrates to ease handling of the 1.2-μm PEN foil for laboratory-scale device fabrication. The PEN film (obtained from Teijin Dupont Films, Japan) adheres to the supporting substrate only by weak van der Waals forces, which readily enables all subsequent processing steps (vacuum evaporation, anodization and chemical vapour deposition). After fabrication, the devices can be peeled off the supporting stack without causing damage. Critical for low-voltage, high-performance OFETs on very thin, ultra-rough foils is a dense, nanometre-thick gate dielectric layer that can be processed at room temperature. Potentiostatic anodization was found to form self-healing nanoscale barrier oxides disregarding the substrate roughness. The technique is readily scalable and complies with industrial ultrathin polymer substrates.

The position b of the neutral mechanical plane (zero-strain position) of a multilayer stack with the nth layer on top of the first layer on the bottom is given by[8]

$$b=\sum_{i=1}^{n} E_i t_i \left[\sum_{j=1}^{i} t_j - \frac{t_i}{2}\right] \Big/ \sum_{i=1}^{n} E_i t_i \qquad (1)$$

where E_i and t_i denote the Young's moduli and thicknesses of the individual layers. For the ultrathin OFETs ($n = 6$; 1.2 μm PEN/25 nm Al/19 nm anodic AlO_x/20 nm DNTT/40 nm Au/0.8 to 1 μm parylene diX-SR) the elastic moduli are $E_{PEN} = 4$ GPa, $E_{Al} = 69$ GPa, $E_{AlO_x} = 130$ GPa, $E_{DNTT} = 2.3$ GPa, $E_{Au} = 79$ GPa and $E_{diX\text{-}SR} = 4$ GPa. This places the neutral mechanical plane at a distance b of 1.15–1.2 μm from the bottom surface of the PEN substrate, near the position of the active OFET structure. Future theoretical work should focus not only on wrinkle frequencies and amplitudes but also on instabilities in wrinkle geometry at large stretch, as well as possible delamination processes of the ultrathin films on elastomeric substrates and heat transport in such multilayer systems.

Full Methods and any associated references are available in the online version of the paper.

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Supplementary Information is available in the online version of the paper.

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Author Information Reprints and permissions information is available at www.nature.com/reprints. The authors declare no competing financial interests. Readers are welcome to comment on the online version of the paper. Correspondence and requests for materials should be addressed to T.So. (someya@ee.t.u-tokyo.ac.jp) or M.K. (martin@ntech.t.u-tokyo.ac.jp).

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METHODS

Ultrathin film substrate. PET sheets (thickness 125 μm) covered with a thin polydimethylsiloxane layer served as supporting substrates to ease handling of the 1.2-μm PEN foil for laboratory-scale device fabrication. The PEN film (obtained from Teijin Dupont Films, Japan) adheres to the supporting substrate only by weak van der Waals forces, which readily enables all subsequent processing steps (vacuum evaporation, anodization and chemical vapour deposition). After fabrication, the devices can be peeled off the supporting stack without causing damage.

Thin-film bolometers, temperature sensors and heating elements. Thin-film bolometers were fabricated by evaporating a resistive layer of aluminium 100 nm thick with 1 mm diameter through a shadow mask onto the 1.2-μm PEN substrate. The bolometers have a small heat capacity $C = 8\ \mu J\ K^{-1}$, and the temperature coefficient of the resistance (TCR) of the aluminium electrode is $0.06\%\ K^{-1}$. Efficient heat transfer from the thin polymer foil to the copper block heat sink is facilitated by a thin layer of a liquid eutectic gallium–indium–tin alloy (galinstan), thus demonstrating efficient heat management strategies vital for high-power plastic electronic devices[31]. The bolometer was connected to a constant-current source delivering a current of 16 mA through the resistor. The resistance change was measured as a voltage decrease in a four-point probe technique configuration[32], and the temporal voltage decrease was recorded directly with a Tektronix TDS 520C oscilloscope after amplification by a factor of 30 with an EG&G 5113 voltage amplifier. The bolometer was calibrated by measuring the TCR of the resistance element on a heating stage. Thermal pulse excitation was performed with a Nd:YAG laser at 1,064 nm with a pulse length of 10 ns and an adjustable pulse energy between 0 and 10 mJ. The transient signals of the bolometer after pulse excitation were fitted (according to equations (1) and (2) in Supplementary Information) with a thermal diffusivity of $D = (7.4 \pm 0.7) \times 10^{-8}\ m^2\ s^{-1}$ (typical of PEN polymers). The temperature prefactor ΔT_0 was adjusted to obtain a best fit to the transients, owing to slight variations in the absorbed energy for the consecutive experiments on free-standing films and films coupled to the heat sink. The experiments show that heat is transferred to the heat sink very rapidly—on a time scale of only $4\tau = 9\ \mu s$ (equation (2) in Supplementary Information). For the thin-film thermometers and resistive heating elements, gold strips 100 nm thick were thermally evaporated through a shadow mask at a pressure of 10^{-6} mbar and a rate of $1\ nm\ s^{-1}$. The strips were 1 mm wide and 1 cm long. Thin copper wires were attached to both ends of the gold strips for four-point resistivity measurements with a Keithley 2400 source meter. Thermal images were recorded by a NEC Thermotracer 7700 with 320 × 240 pixel resolution.

Thin-film transistors. The gate electrode was prepared by thermal evaporation of 40 -nm aluminium through a shadow mask at a pressure of 10^{-6} mbar and a rate of $10\ nm\ s^{-1}$. The high-k aluminium oxide gate dielectric layer was formed by potentiostatic anodization under ambient conditions. A 0.01 M citric acid monohydrate ($C_6H_8O_7 \cdot H_2O$) electrolyte was prepared with ultrapure (18 MΩ cm) water. The contacted gates were then immersed in the electrolyte to form the working electrode (anode), and a platinum foil served as the counterelectrode. The thickness of the anodic oxide d_{ox} was determined by the applied anodization potential U according to $d_{ox} = k(U - U_{ox})$. The oxide formation factor k is $1.6\ nm\ V^{-1}$ for aluminium[17,18]. U_{ox} accounts for the naturally occurring passivation layer, which is about 2 nm, or $U_{ox} = -1.35$ V for aluminium. Anodic oxide formation on valve metals is a self-inhibiting process; thus, it is self-healing and converges to a uniform film thickness for a given potential. Optimal oxide quality is obtained when the anodization current drops to a steady-state value of about 1 μA for our geometry, and is typically reached within 10 min. Because the liquid electrolyte readily wets even a rough, bumpy surface, electrochemical oxidation is a highly reliable method of forming a dense dielectric layer, which ultimately decreases leakage currents and enables high device yields. To demonstrate the precise scalability of the dielectric thickness and thus the operating voltage of the transistor, devices were fabricated with oxide 9 nm and 19 nm thick (operated at up to 3 V and up to 5 V, respectively). We functionalized the aluminium oxide surface with an n-tetradecyl phosphonic acid SAM[33] to form a suitable interface to the organic semiconductor DNTT and decrease the interface trap density. After brief surface activation in oxygen plasma (1 min at 100 W), the gates were dipped into a 5 mM solution of the SAM molecules in propan-2-ol to form a SAM 2 nm thick. Our hybrid gate dielectric layer has interface capacities of about $420\ nF\ cm^{-2}$ and about $280\ nF\ cm^{-2}$ for devices with 9 nm alumina and 19 nm alumina, respectively. Subsequently, 20 nm DNTT was evaporated through a shadow mask at a pressure of 10^{-6} mbar and a rate of $0.1\ Å\ s^{-1}$. Gold 30 nm thick evaporated through a shadow mask at a pressure of 10^{-6} mbar and a rate of $1\ Å\ s^{-1}$ formed source and drain electrodes with width 500 μm and length 40 μm for single transistors. We note that our transistor fabrication process is compatible with n-type or ambipolar organic semiconductors[34–37], which may further expand possible applications of imperceptible electronics by taking full advantage of organic complementary circuits and device architectures[6,38–40].

Active-matrix tactile sensing foil. The fabrication methods employed for a single transistor are readily scalable and require no additional process steps for the 12 × 12 active-matrix transistor backplane. Transistors with a channel width of 12,000 μm and length of 20 μm provide high on-currents at low operating voltage. For the active-matrix transistors, 19-nm anodic alumina was formed. We deposited parylene diX-SR (about 800 nm thick; Daisan Kasei Co. Ltd) by chemical vapour deposition to separate the switching matrix backplane from the sensor layer. A green laser marker system (T-Centric SHG Laser Marker; Keyence) was used to drill precise, high-resolution vias to the underlying transistors (the typical via diameter was 30 μm). Interdigitated gold electrodes 100 nm thick with a 30 μm pitch were evaporated through a shadow mask at a pressure of 10^{-6} mbar and a rate of $1\ Å\ s^{-1}$, and served as resistive tactile sensors. To perceive pressure signals from non-conducting objects, a pressure-sensitive rubber sheet (CS57-7RSC; PCR Technical Corp.) was laminated on top of the tactile foil.

Neutral mechanical plane calculation. The position b of the neutral mechanical plane (zero-strain position) of a multilayer stack with the nth layer on top of the first layer on the bottom is given by[8]

$$b = \sum_{i=1}^{n} E_i t_i \left[\sum_{j=1}^{i} t_j - \frac{t_i}{2} \right] \Big/ \sum_{i=1}^{n} E_i t_i \qquad (1)$$

where E_i and t_i denote the Young's moduli and thicknesses of the individual layers. For the ultrathin OFETs ($n = 6$; 1.2 μm PEN/25 nm Al/19 nm anodic AlO_x/20 nm DNTT/40 nm Au/0.8 to 1 μm parylene diX-SR) the elastic moduli are $E_{PEN} = 4$ GPa, $E_{Al} = 69$ GPa, $E_{AlO_x} = 130$ GPa, $E_{DNTT} = 2.3$ GPa, $E_{Au} = 79$ GPa and $E_{diX\text{-}SR} = 4$ GPa. This places the neutral mechanical plane at a distance b of 1.15–1.2 μm from the bottom surface of the PEN substrate, near the position of the active OFET structure.

Compression and stretching. Transistors for the compression/stretching experiments were fabricated as described above for single transistors but with extended gate, source and drain contacts to enable electrical connection for the bending and stretching tests. The devices were encapsulated with a parylene layer 0.8–1 μm thick that placed the transistors near the neutral plane and protected them from mechanical damage. Transistors for the stretching/compression tests had a channel width of 1,000 μm and length of 100 μm. A schematic diagram of the compression/stretching setup together with a picture of the setup is shown in Supplementary Fig. 17. A strip of 3M VHB 4905 elastomer was fixed between two clamps and pre-stretched. Polyimide tape was used as a rigid delimiter to define the compression and stretching zone, and the distance between the two rigid delimiters was measured to be exactly 1 cm (defined as L_0). The transistors were then placed on the elastomer. The active device area (gate area) of the transistor was $1.75 \times 1.5\ mm^2$. Together with the gate, source and drain contacts, the transistor occupied roughly 65% of the 1 cm gap in the compression direction. The transistor was then compressed by moving the rigid delimiters closer together, thereby allowing the elastomer to relax, and compressing the device area. The compression zone measured $2 \times 1\ cm^2$, meaning that the deformation was quasi-linear and quasi-reversible, although there was always a small, unintentional biaxial and/or permanent deformation due to the viscoelasticity of the elastomer. We defined the compression as $(L_0 - L)/L_0$. Shrinking the gap from 10 mm to 3 mm corresponded to a 70% compression. The definition of tensile strain used the fully relaxed (smallest) state as the starting point. After relaxation to 3 mm, re-stretching to 10 mm gave a tensile strain of (10–3 mm)/3 mm; that is, 233% tensile strain. For the biaxially stretchable transistors, the elastomer was fixed at all sides and pre-strained by pressing a 1-cm cylinder against the bottom surface. The transistor foil was then applied to the top surface and the cylinder was lowered, relaxing the elastomer. The change in device area was measured by monitoring circular markers on the elastomer.

Device characterization. Electrical characterization of transistors and active-matrix sensing sheets was performed with a Semiconductor Parameter Analyser (Agilent B1500A) under ambient laboratory conditions. Thermal stability tests were performed under ambient atmospheric conditions. Immersion tests in aqueous solution were performed in saline solution (0.9wt% NaCl; Becton Dickinson and Company prepared saline solution, normal (physiological)). Optical micrographs and three-dimensional topological images were recorded with a Keyence VK-9710 ultraviolet laser microscope and a Keyence VHX-2000 digital microscope. SEM images of wrinkled transistors on the elastic substrate were taken with a Hitachi S-4800 Field Emission Scanning Electron Microscope at an acceleration voltage of 20 kV after fixing the samples onto a rigid glass substrate and depositing a gold layer less than 5 nm thick. Cross-sectional SEM (S-5200 Ultra-High Resolution FE SEM) and high-resolution STEM (200-kV HD-2700 Cs-corrected STEM) images were taken by Hitachi High-Technologies Corporation. Surface morphologies were characterized with a NanoScope IIIa (Veeco Instruments) atomic force microscope.

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